

Voltaro — battery swapping for commercial two-wheelers

The Problem

A delivery rider on an electric scooter loses between forty minutes and two hours a shift to charging. That is unpaid time, and it is the single largest reason riders in emerging markets stay on petrol despite fuel costing them a third of their earnings.

Charging infrastructure does not solve this. A rider cannot wait, and a battery large enough to avoid waiting makes the vehicle too expensive to buy.

Our Solution

Voltaro operates a network of automated battery swap stations. The rider pulls in, exchanges a depleted battery for a charged one, and is moving again in under ninety seconds.

The batteries are ours and the rider never owns one, which removes the largest single cost from the vehicle purchase and removes battery degradation risk from the rider entirely. Our station is the size of a vending machine, requires a standard grid connection, and holds sixteen batteries.

The battery management system is our own, and it is genuinely good: we track cell-level health across the fleet and retire packs before they fail rather than after.

The Market

There are approximately 290 million two-wheelers in the markets we are targeting, of which an estimated 24 million are used commercially for delivery and ride-hailing. Fuel and maintenance spend across that commercial fleet is on the order of \$34B a year.

Electrification of this fleet is not a question of whether. Three of our target markets have announced dates by which new commercial two-wheeler registrations must be electric.

This is one of the largest addressable markets in mobility and it is almost entirely unserved.

Competition

Two companies operate swap networks at scale in Asia, neither in our markets. Several vehicle manufacturers have announced swap-compatible models without building networks.

The network is the hard part and the defensible part. Station density is what makes the service usable, and density is expensive to reach and expensive to attack.

The Team

Kwame Asante spent seven years in energy infrastructure, including three deploying distributed solar. Ingrid Lohse was a battery systems engineer at an automotive supplier.

Thirty-eight people.

Traction

We have stations deployed and riders using them.

Adoption has been strong since launch, the network is growing month on month, and rider feedback has been overwhelmingly positive. Utilisation is healthy and improving as density increases in our pilot city, and we are seeing the network effects we expected: each new station makes the existing ones more useful.

We have interest from several large delivery operators and expect to convert some of those conversations into fleet agreements.

Business Model

Riders subscribe monthly for unlimited swaps, or pay per swap. The subscription is priced below what the rider currently spends on fuel, which is what makes the decision easy.

Unit economics at station level are attractive once utilisation matures.

Go-To-Market

Fleet agreements with delivery platforms, plus direct rider acquisition at the stations themselves.

Funding

We are raising \$18M to expand the network, grow the battery fleet, and enter a second city.